

Dot Product. Given two vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$,

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} &= u_1 v_1 + u_2 v_2 + u_3 v_3 \\ &= |\mathbf{u}| |\mathbf{v}| \cos \theta\end{aligned}$$

Orthogonal Projection of $\mathbf{u}$ onto $\mathbf{v}$,

$$\begin{aligned}\text{proj}_{\mathbf{v}} \mathbf{u} &= |\mathbf{u}| \cos \theta \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right) \\ &= \text{scal}_{\mathbf{v}} \mathbf{u} \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right) \\ &= \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \right) \mathbf{v}.\end{aligned}$$

Scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$,

$$\text{scal}_{\mathbf{v}} \mathbf{u} = |\mathbf{u}| \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}.$$

Work. Let $\mathbf{F}$ be a constant force applied to object, producing a displacement $\mathbf{d}$. If the angle between $\mathbf{F}$ and $\mathbf{d}$ is θ , then the **work** done by the force is

$$W = |\mathbf{F}| |\mathbf{d}| \cos \theta = \mathbf{F} \cdot \mathbf{d}.$$

Cross product, of two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$,

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta$$

$$|\tau| = |\mathbf{r}| |\mathbf{F}| \sin \theta$$

$$\mathbf{u} \times \mathbf{v} = \det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \det \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \det \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \det \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}.$$

Helpful formulas

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain rule)
3. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx}e^x = e^x$
5. $\frac{d}{dx} \sin ax = a \cos ax$
6. $\frac{d}{dx} \cos ax = -a \sin ax$
7. $\int x^n dx = \frac{1}{n+1} x^{n+1} + C; n \neq -1$
8. $\int \cos ax dx = \frac{1}{a} \sin ax + C$
9. $\int \sin ax dx = -\frac{1}{a} \cos ax + C$
10. $\int \cos^2 x dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$
11. $\int \sin^2 x dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$

